

components and functionality:

1. **Arduino Uno:** The main microcontroller used to control the system.
2. **Ultrasonic Sensor (HC-SR04):** Measures the water level by emitting sound waves and detecting the time it takes for the echo to return.
3. **Relay Module:** Used to control the water pump automatically based on the water level.
4. **Water Pump:** Likely a DC pump controlled by the relay to fill the water tank when required.
5. **LCD Display (16x2):** Displays real-time water levels or system status.
6. **Battery and Power Circuit:** Provides power to the system, possibly with a 2000mAh Li-ion battery visible in the image.
7. **LED Indicators:** Provide visual feedback about the system's operational status.
8. **Perforated PCB with a Potentiometer or Sensor:** Might be used for additional control or sensor calibration.

Components and Their Roles

1. **Arduino Uno (Microcontroller):**
 - Acts as the brain of the system.
 - Reads sensor data, processes it, and sends control signals to the relay module to operate the water pump.
2. **Ultrasonic Sensor (HC-SR04):**
 - Measures the distance from the sensor to the water surface inside the tank.
 - Calculates water level based on the tank's dimensions and sends this data to the Arduino.
3. **Relay Module:**
 - Electrically isolates the Arduino and controls the high-power water pump based on signals from the Arduino.
 - Turns the pump ON/OFF depending on the water level.
4. **Water Pump:**
 - Fills the tank with water when the Arduino detects a low water level.
 - Stops when the tank reaches the predefined "full" level.
5. **LCD Display:**
 - Displays the current water level and the pump's operational status (e.g., "Pump ON," "Tank Full").
6. **Power Supply (Battery + Custom Circuit):**
 - The battery provides power to the Arduino, sensors, and pump. The custom circuit ensures stable power delivery and may filter noise or manage voltage levels.
7. **LED Indicators:**
 - Provide visual cues for different statuses, such as when the pump is running or when the system is powered on.

Working of the System

1. **Startup:**
 - The Arduino initializes the system components.
 - The ultrasonic sensor begins measuring the distance to the water surface.
2. **Water Level Detection:**
 - The ultrasonic sensor sends distance data to the Arduino.
 - The Arduino calculates the water level based on the tank's height and the measured distance.
3. **Pump Control:**
 - **If the water level is below the minimum threshold:**
 - The Arduino sends a signal to activate the relay, turning the pump ON.
 - **If the water level reaches the maximum threshold:**
 - The Arduino deactivates the relay, turning the pump OFF.
4. **Display Updates:**
 - The current water level, pump status, or any alerts are shown on the LCD.
5. **Power Management:**
 - The battery powers the system, while the custom circuit may stabilize or step down the voltage for consistent operation.

Applications

- Household water tanks.
- Agricultural water storage and irrigation systems.
- Industrial water level monitoring.

What the Code Does

This program uses an **Arduino** to monitor the water level in a tank and control a water pump and buzzer accordingly. It also displays the water level on an **LCD screen**.

How It Works

1. **Initialization (setup):**
 - The LCD, water pump, buzzer, and ultrasonic sensor are set up.
 - The motor is turned **off** initially.
 - A message "Water Level Indicator" is displayed on the LCD for 2 seconds.

2. Measuring Water Level:

- The **ultrasonic sensor** (trigPin and echoPin) measures the distance to the water surface.
- The distance is calculated in **centimeters** and shown on the LCD.

3. Decision Making (loop):

Based on the water level (distanceCm), the program controls the motor and buzzer:

- **Full Tank (distance $\leq$ 4 cm):**
 - Buzzer turns **ON** briefly.
 - Motor stays **OFF** (water filling stops).
 - LCD shows: "WaterLevel:Full" and "Motor:Off".
- **Medium Level (distance between 12 and 15 cm):**
 - Motor remains **OFF**.
 - LCD shows: "WaterLevel:Med" and "Motor:Off".
- **Empty/Low Level (distance > 28 cm):**
 - Motor turns **ON** to fill the tank.
 - LCD shows: "WaterLevel:Empty" and "Motor:ON".

4. Loop Repeat:

The program continuously repeats this process, updating the water level and controlling the pump accordingly.

Pin Assignments

1. LCD Pins

The **Liquid Crystal Display (LCD)** uses 6 pins connected to Arduino for communication:

```
const int rs = A0, en = A1, d4 = A2, d5 = A3, d6 = A4, d7 = A5;
```

- **A0** → **rs** (Register Select): Controls whether data is being sent as **commands** or **characters**.
 - **A1** → **en** (Enable): Enables the LCD to receive data.
 - **A2 to A5** → **d4, d5, d6, d7** (Data Pins): These are used to send 4-bit data to the LCD.
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2. Water Motor Pin

```
const int wat_motor = 8;
```

- **Pin 8** controls the **water pump motor** through a relay module.
 - **HIGH** → Motor OFF (Relay inactive).
 - **LOW** → Motor ON (Relay active).
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3. Buzzer Pin

```
const int buzzer = 9;
```

- **Pin 9** is connected to the **buzzer** to alert when the water tank is full.
 - **HIGH** → Buzzer ON.
 - **LOW** → Buzzer OFF.
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4. Ultrasonic Sensor Pins

```
const int trigPin = 10;
```

```
const int echoPin = 11;
```

- **Pin 10** → **trigPin** (Trigger): Sends a **10-microsecond pulse** to the ultrasonic sensor to start the measurement.
 - **Pin 11** → **echoPin** (Echo): Receives the pulse bounced back from the water surface. The duration is measured to calculate the distance.
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Pin Connections Summary

Component	Arduino Pin	Purpose
LCD RS (Register Select)	A0	Select command/data mode
LCD EN (Enable)	A1	Enable communication
LCD Data Pins (d4–d7)	A2, A3, A4, A5	Send 4-bit data to LCD
Water Pump Motor	8	Turns the motor ON/OFF
Buzzer	9	Alerts when tank is full
Ultrasonic Trig Pin	10	Sends trigger pulse
Ultrasonic Echo Pin	11	Receives echo pulse

Power Pins (Assumed)

- **VCC:**
 - Powers the LCD, Ultrasonic Sensor, Relay Module, and Buzzer.
- **GND:**
 - Common ground for all components.

Analog Pins

- **A0 to A5** are used to connect the **LCD**.
 - These pins are used as **digital output pins** in this project.
 - Arduino analog pins can also function as digital pins when configured in the code.
 - **Purpose:** Sending data and commands to the LCD for displaying water level status.
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Digital Pins

- The following **digital pins** are used:
 - **Pin 8** → Controls the **water motor** (Relay control).
 - **Pin 9** → Controls the **buzzer** for alert.
 - **Pin 10** → **trigPin** of the ultrasonic sensor to send the trigger pulse.
 - **Pin 11** → **echoPin** of the ultrasonic sensor to receive the reflected pulse.